SECTION 14.0

OFFICE-BASED ANESTHESIA

PROCEDURAL SPECIALISTS

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ANESTHESIOLOGISTS

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INTRODUCTION—ANESTHESIOLOGIST'S PERSPECTIVE

Richard A. Jaffe, MD, PhD and Terri D. Homer, MD

DEFINITION OF OFFICE-BASED ANESTHESIA

Office-based anesthesia (OBA) is distinct from outpatient anesthesia in a freestanding surgery facility. OBA is used in many medical specialties, including most dental subspecialties, dermatology, plastic surgery, ophthalmology, otolaryngology, and gynecology. The procedures described in this chapter are representative of the ways in which anesthesia is used in medical/dental offices. Although anesthesia practiced in an office carries the same risks, burdens of responsibility, and skill requirements as in a fully equipped surgical center, in the office setting, the anesthesiologist may be expected to arrange for the oxygen supply, suction, monitoring, and emergency equipment. They may even bring a portable anesthesia machine with them from site to site. This unique challenge for the anesthesia provider includes working with personnel unfamiliar with anesthesia concerns, converting the office into a facility appropriate for anesthesia, selecting appropriate patients, providing safe and effective anesthesia/analgesia, and properly preparing and recovering the patients.

STATE REGULATIONS REGARDING OFFICE-BASED ANESTHESIA

Many states have laws listing strict requirements for medical facilities where anesthesia is provided for surgical procedures. Some states have regulations based on the type of surgical procedure performed. Others regulate and credential the facility based on the type of anesthesia used (i.e., GA, iv, local). Still others base regulations on the type of facility itself. In California, for example, dental offices are regulated differently than medical offices. For many years, the California Dental Board has regulated anesthesia in the dental or oral surgery office by credentialing the anesthesia provider and/or the office facility itself. An oral surgeon, dentist, or physician issued a GA permit by the Dental Board goes through a credentialing process by one or two examiners that includes direct observation of an anesthesia case, demonstration of emergency drills, and examination of required monitoring and resuscitation equipment on site. The permit allows the holder to provide GA in any dental office. The Dental Board also issues "Conscious Sedation" permits to those dental practitioners who qualify and want to use this technique. Physicians in California wishing to have iv sedation or general anesthesia available in their office must first be accredited as a surgery center by one of several agencies providing this service such as AAAHC.org.

EQUIPMENT NEEDED FOR OFFICE-BASED ANESTHESIA

In the office setting, the anesthesia provider may or may not have the use of an anesthesia machine or other sophisticated equipment that is readily available in a surgical center. In some respects, however, this setting is analogous to other non-OR locations. The ASA Guidelines for Nonoperating Room Anesthetizing Locations covers all types of out-of-OR facilities, and these recommendations should be followed. Appropriate monitoring—including pulse oximetry, ECG, and BP—is required. Many portable monitors have $ETCO_2$ monitoring capability, which may be useful for the spontaneously ventilating, sedated patient. A precordial stethoscope is quite useful for monitoring respirations, especially in the dental patient, in whom airway obstruction is a frequent occurrence during the procedure. Also in accordance with the ASA guidelines, full resuscitation equipment, an adequate source of O_2 (and backup O_2), a functioning suction, adequate lighting and electrical outlets (with backup battery source), and a telephone with immediate access to a hospital ER also must be available. In the credentialed medical (nondental) facility, these items are required to be on site. In the dental facility, they may or may not be present. It is the responsibility of the anesthesia provider to make sure these items are available in the medical or dental facility before administering an anesthetic.

ANESTHETIC TECHNIQUES IN THE OFFICE SETTING

Typically, the anesthetic techniques in the office setting can range from conscious sedation to GETA. The type of procedure, the patient's clinical status, and the surgeon or dentist's preference all contribute to the decision of which technique is best for that particular situation. In the pediatric patient having dental restoration work, "conscious sedation" is not adequate because most often the dental restoration required is very extensive and will likely take several hours to accomplish. In this situation, GETA with nasal intubation (or oral intubation if nasal intubation is not possible) is usually the best choice of anesthetic technique. In the patient having dental implant placement, minimal or moderate sedation may be all that is required. The oral surgeon may require the patient's cooperation at times during the procedure. The patient undergoing full-face laser resurfacing will most likely require deep sedation or GA because this procedure can be quite painful. However, because sedation is a continuum, the anesthesia provider must be prepared to rescue any patient receiving any sedation in the office (see Continuum of Depth of Sedation, ASA Guidelines).

PATIENT SELECTION

As in all medical facilities, the patient's safety is of paramount importance. In the office setting, the ability to achieve a successful outcome is dependent first of all on appropriate patient selection. The patients presenting for OBA will fall into several categories, depending on the procedure and the patient's age and medical condition. An "appropriate patient" can be an ASA 1 or 2 patient. They may even be an ASA 3 if (a) their medical problems are stable and well controlled with medication and (b) the office procedure itself will not pose an undue risk to them.

PREOPERATIVE PREPARATION

An important role of the anesthesiologist is to educate the patient (or patient's parents, as appropriate) about the office anesthesia experience. A preop phone call discussing the patient's medical history, past anesthesia experience (in a hospital, surgery center, or dental office), the NPO requirements, the anesthesia technique(s) to be used, postanesthesia expectations, and the anesthesia fees is essential. A written packet describing some of this information can be given to the patient in advance.

Safe and accepted NPO requirements on the day of the procedure are as follows:

- A light meal at least 6 h before the appointment.
- Clear liquids (including Gatorade, Jell-O, fruit popsicles) up to 2 h before the appointment.
- The patient's usual medications should be continued on the day of the procedure.

RECOVERY AND DISCHARGE

In a medical or dental office, often there is no separate recovery area designated as such. It is common practice for the patient to be recovered by the anesthesiologist in the treatment room until they can open their eyes and maintain an adequate airway without assistance. At that point, any iv or monitors that may have been used can be removed. If it is a pediatric patient, the parents may be brought into the room, although recovery remains under the supervision of the anesthesiologist. Office anesthesia patients can be discharged when they are well oriented, their pain and nausea are controlled, and they have a responsible adult to accompany them. They may still feel drowsy, but this should not prevent them from being able to walk with assistance.

Discharge instructions regarding appropriate postop activities should be given to the responsible adult with the patient. Generally, patients are asked to adhere to the following instructions upon discharge:

- NPO except clear liquids in the first 2 h after arriving at home. (Unless the ride is >1

h, we ask the patient not to drink anything in the car on the way home from the procedure facility.)
- A light meal after the first 2 h, if the patient wishes.
- Adults should take it easy the rest of the day and have a responsible adult companion for at least 4 h after the procedure.
- No driving for 24 h.
- Children should stay home for the rest of the day postprocedure, under the direct supervision of a responsible adult.

The anesthesiologist also should give the responsible party (parent, friend, other relative) his/her pager or cell phone number in the event they need to contact the anesthesiologist after patient discharge.

Suggested Readings

1. American Society of Anesthesiologists: Office Based Anesthesia and Denta Anesthesia guidelines 2019: https://www.asahq.org/advocacy-and-asapac/advocacy-topics/office-based-anesthesia-and-dental-anesthesia
2. Lieblich S: Preoperative evaluation and patient selection for office-based oral surgery anesthesia. Oral Maxillofac Surg Clin North Am 2018; 30(2):137–44.
3. Schwartz A: Airway management for the oral surgery patient. Oral Maxillofac Surg Clin North Am 2018; 30(2):207–26.
4. Shapiro FE, Punwani N, Rosenberg NM, et al: Office-based anesthesia: safety an outcomes. Anesth Analg 2014; 119(2):276–85.
5. Voulgarelis S, Scott JP: Monitoring for nonoperating room anesthesia. Anesthesiol Clin 2017; 35(4):591–9.

FACIAL REJUVENATION: LASERS AND RF TISSUE TIGHTENING

SURGICAL CONSIDERATIONS

David A. Berman, MD

Description

A variety of medical devices have been approved for use to improve skin concerns, such as wrinkles, precancerous skin lesions (actinic keratoses), discoloration, acne scars, traumatic scars, and sagging skin. In the mid-1990s, laser resurfacing using the fully ablative CO_2 laser was very popular. This procedure yielded wonderful results, but extensive heat from this procedure caused some undesirable side effects such as scarring or pigmentation problems, prompting the industry to develop the cooler and more conservative resurfacing tool using the erbium laser. Less than a decade later, a novel fractionated laser resurfacing device was developed that removed only a fraction of the skin in a pattern reminiscent of pixels in newspaper print. Certainly, the risks and side effects were reduced compared to the older technology, though this more conservative approach rendered a more modest improvement of skin concerns. Further research led to the development of nonablative technologies such as radiofrequency/ultrasound/nonlaser light-based devices, most of which resulted in better side effect profiles and positive outcomes. Although the holy grail of devices would render a no downtime, painless procedure void of negative effects, the search for such continues to elude us. Meanwhile, physicians who perform these procedures should encourage patients to opt for office-based anesthesia and thus make the experience a more pleasant one for all involved.

Because some devices deliver a tremendous amount of heat to the skin surface, ablating both the epidermis and a portion of the dermis, nerve blocks and local anesthetic infiltration are often inadequate, thus requiring either iv sedation or GA. Facial nerve blocks may be performed to supplement iv sedation. After a Betadine® prep, anesthetic eye drops are used, followed by the insertion of protective corneal shields. The treatment then begins with one or more passes performed at various energy levels.

Usual preop diagnosis

Wrinkles; precancerous skin lesions (actinic keratoses); acne scars; traumatic scars

SUMMARY OF PROCEDURE	
Position	Supine, with shoulder roll. Surgeon at head of table
Antibiotics	None during procedure. Some patients will require antivirals and/or antibiotics started 1 d before procedure.
Surgical time	30–60 min
Postop care	Supine position, occasional ice packs, antiviral medications/antibiotics. Some require facial dressings.
Mortality	None reported
Morbidity	Bacterial infection Herpes simplex virus reactivation Delayed reepithelialization

	Fat atrophy Hypertrophic or keloid scar formation Corneal abrasion Delayed healing with redness Hyperpigmentation and hypopigmentation
Pain Score	3–6 depending on the type of medical device used

PATIENT POPULATION CHARACTERISTICS	
Age range	18–85 yr
Male:Female	1:4
Incidence	*Facial rejuvenation procedures: ~340,000/yr in US
Etiology	Solar radiation, acne, or trauma resulting in scars

*2018, American Society for Aesthetic Plastic Surgery.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Some facial rejuvenation procedures are quite painful and can be very stressful to the patient; thus, local anesthesia is often inadequate. For these procedures, it is important to make sure that the patient with chronic HTN and/or other cardiovascular or respiratory disease is being adequately treated for these problems before undergoing either laser resurfacing or RF tissue tightening. A preop ECG (taken within the last year) is recommended for those patients >60 yr old or those being treated for HTN.

Premedication	In the adult patient, an oral premed can be given, if necessary, before the patient arrives in the office for the procedure. Lorazepam 0.5–1.0 mg po 1 h before the appointment will help relax the severely anxious patient.

INTRAOPERATIVE

Anesthetic technique

MAC and/or GA

MAC/GA	During full-face laser resurfacing procedures, the patient will need to be under deep iv sedation, unless the surgeon is willing to use a large amount of local anesthetic (often RF tissue tightening will require less sedation, depending on the device used). Usually, a combination of intermittent doses of midazolam (1–2 mg iv), meperidine (25 mg iv), ketamine (25–30 mg iv q1h prn), and propofol (25–50 mcg/kg/min iv) will provide an adequate anesthetic state. If a nasal airway is inserted, nasal O_2 can be used (limit FiO_2 to ≤0.30). Otherwise, O_2 should be administered between facial passes if a laser is used. Again, the patient is not electively intubated, but an LMA may be needed if it is difficult to maintain a patent airway. Usually, patients having this procedure will require at least 75–150 mg meperidine (or its equivalent) to relieve the painful burning caused by multiple passes of these devices. Patients will shiver and sustain HTN and tachycardia unless an adequate level of analgesia is achieved. Because of the narcotic requirement, they should have prophylactic antiemetics during the procedure. Dexamethasone (8 mg iv) and metoclopramide (15 mg iv) or ondansetron (4 mg) are a good combination.	
Blood and fluid requirements	IV: 20 ga × 1 BSS @ 2–4 mL/kg/h	
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes (eye shields in place)	
Complications	Laser fire	Use intermittent, low-flow O_2. Consider wrapping foil around cannula. Limit FiO_2 to ≤0.30.

POSTOPERATIVE

Complications	PONV
Pain management	Oral analgesics
Recovery and discharge	The patient recovering from a full-face procedure usually has received a lot of anesthetic medication in a short period of time. They often require at least 45–60 min to recover in the office. Vital signs should be stable (↑ BP and HR must be treated). Pain and nausea also should be under control. These patients must be accompanied by a responsible adult who can stay with them at home for a few hours after the procedure. A follow-up phone call to the patient by the anesthesiologist that evening is very helpful and appreciated.

OFFICE DENTAL REHABILITATION UNDER DEEP IV SEDATION

SURGICAL CONSIDERATIONS

Vernon J. Adams Jr, DMD

Description

Dental rehabilitation includes the restoration of good dental health by removal of caries and decayed teeth, replacement of crowns or bridges, root canals, and periodontal treatment. A throat pack with external tie is placed and a mouth prop inserted. Dental rehabilitation is usually done in phases, one quadrant at a time. A rubber dam is placed to isolate the teeth that will be treated. All dental caries are removed, the restorations are placed, and all debris is irrigated and suctioned away from the rubber dam, which is then removed. This sequence is repeated in other quadrants as needed. After the operative phase of treatment is completed, the next treatment phases proceed (e.g., taking impressions for various appliances or dental prosthetics, dental extractions, or a dental prophylaxis). After the treatment is completed, the dentist examines the mouth and removes any loose gauze, debris, or other material left over from the treatment before the anesthesiologist begins the process of emergence from anesthesia.

Usual preop diagnosis

In-office dental rehabilitation usually focuses on treatment of the patient who is unable to be treated in a conventional setting for primarily behavioral/emotional reasons. These include dentophobia; pediatric patients who are excessively apprehensive and/or combative or who have a significant amount of dental caries; “special needs” pediatric patients—autistic, mentally retarded, developmentally delayed—and patients with mild-to-moderate forms of cerebral palsy.

SUMMARY OF PROCEDURE	
Position	Supine, on a dental chair or bench. A pillow under the knees will improve comfort and prevent sliding.
Incision	± Gingival or intraoral mucosa
Special instrumentation	Dental setup
Unique considerations	Intubation recommended; throat pack placed; airway protection of paramount importance
Antibiotics	None used routinely
Surgical time	90–180 min
EBL	Minimal
Postop care	Recover in lateral position
Mortality	Rare
Morbidity	Biting of lips or tongue 2° local anesthesia: 1% Bleeding: 1% Infection: <1% Delayed bleeding: rare
Pain Score	1–3; higher for certain procedures (e.g., extraction of impacted tooth or when all quadrants of the mouth are restored)

PATIENT POPULATION CHARACTERISTICS	
Age range	13 mo+
Male:Female	1:1
Incidence	>20,000/yr in the United States
Etiology	Poor oral hygiene and/or dietary habits; lack of

	continuing periodic dental care

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

In the pediatric dental office, the patients presenting for GA are those who cannot cooperate due to age (too young), too anxious, or to a preexisting mental or physical disease. Examples include a 20-mo-old child with early childhood caries, an 8-yr-old autistic child, or a child of any age with cerebral palsy, obesity, Down syndrome, or other congenital syndrome. Obviously, the anesthetic treatment plan must be tailored to the patient's individual needs. These patients must be screened in advance for clinical conditions that would put them at risk for problems intraop or postop. Specifically, a patient with any cardiac, respiratory, endocrine, or neurologic problem must be evaluated. If the clinical condition is mild, stable, and under good control, the patient may be considered for GA in the office. Examples of such conditions include mild non–steroid-dependent asthma, corrected congenital heart disease, or a stable Sz disorder. The child with Down syndrome, obesity, snoring, or OSA may pose a very difficult problem in this setting because airway obstruction can occur easily under deep sedation. Also, these patients frequently have concurrent congenital heart disease. The child with a current or recent URI always poses a dilemma. Although there is controversy on this issue, these children generally should have their procedures postponed because their chances of sustaining periop respiratory problems are higher than normal.

Premedication	There are several options available for preprocedure sedation. In the older child (>8 yr) who is psychologically mature, an iv can be placed without premedication. In a dental office that is equipped with N_2O, the patient can breathe a high-flow N_2O/O_2 mix through a nasal mask for 10 min before iv placement. If the patient accepts the mask, this technique can help lessen the patient's fear of and reaction to the iv. The use of a 30-ga needle for infiltration of buffered 2% lidocaine and, sometimes, the use of EMLA cream (placed 30 min in advance) can be very helpful. Oral premedication is used commonly in children 4–8 yr old. Midazolam (0.5–0.7 mg/kg; maximum = 20

mg) mixed in an appropriate dose of liquid acetaminophen, given in the office 20 min before the procedure, can provide adequate sedation. In some, the addition of oral ketamine 2–3 mg/kg to the midazolam solution may result in a better sedating effect, especially in older pediatric patients.

Although an oral premed is popular and usually effective in the hospital setting, it may not be appropriate in the setting of the dental office for a variety of reasons. First, many of these children are not cooperative with oral hygiene or with taking oral medications at home and will not cooperate with taking them in the office. The special-needs children or adults will often be totally unable to take an oral premed. Second, many of the patients are anxious that the oral premed may not be effective. Third and may be most important, having any volume in the stomach adds to the risk of aspiration management, which is extremely problematic in the dental office.

One safer and certainly more effective and predictable premedication for the young healthy child (6 yr or younger) or for the special-needs child is an intramuscular injection consisting of ketamine 2 mg/kg, midazolam 0.2 mg/kg, and glycopyrrolate 0.1 mg/kg. The anesthesiologist can administer this injection while the patient is in his or her parents' arms in the waiting area. Within 5 min after the injection, the patient will be sedated adequately to allow a smooth separation from the parent and sufficient amnesia for induction. Most parents will accept this premed option when the anesthesiologist explains the rationale for it in advance, usually during the preop phone call.

In the child over age 6, an inhalation induction without premed may work very well, especially when explained properly to the child and the parent.

After the premed, the induction of GA in the adult, pediatric, or special-needs patient in the dental office is similar to anesthetic induction in the OR. Monitors are placed, and in the pediatric patient, this is followed by an inhalation induction, and then the placement of the iv. Intubation (preferably a nasal ETT) follows. After confirmation of proper ETT placement, throat packs are placed with dental floss tied to them and left outside the mouth as a reminder of their presence in the throat. If x-rays are needed, the dentist and/or the dental assistants will get these done after the patient is intubated and then, finally, dental treatment can begin.

There are no special considerations for maintenance of GA, and an inhalation agent, TIVA or a combination, can be used. Some intraop narcotic should be used in the pediatric patient for postop analgesia as well as to minimize emergence delirium. Local anesthetic is often avoided in the pediatric patient undergoing a dental procedure under GA to avoid the possibility of the child biting a numb lip, cheek, or tongue as they recover. In the healthy adult patient undergoing dental work or oral surgery with GA, local anesthesia is usually administered by the dentist or oral surgeon intraop. The routine use of prophylactic intraop antiemetics is highly recommended and may include a combination of dexamethasone, metoclopramide, and ondansetron.

INTRAOPERATIVE

Anesthetic technique

If necessary, a flexible LMA can be used for airway management. Following sedation, the patient is placed in the dental chair, with the head positioned to maintain an open airway. A shoulder roll may be needed to help with head extension. Monitoring—including pulse oximeter, ECG, BP cuff, and precordial stethoscope—is attached. O_2 is supplied, and $ETCO_2$ can be measured through a nasal cannula. An iv is started if not previously placed. A nasal airway is positioned with care (to avoid epistaxis) after the dental x-rays are taken. The anesthesiologist must be constantly vigilant in maintaining an open airway in these patients in the face of an oral procedure. Placing a flexible LMA may make airway management easier without interfering with the dental procedure. A propofol infusion can be started at a rate of 100–150 mcg/kg/min. Administration of a small amount of meperidine (e.g., 5–15 mg iv) may be useful in alleviating emergence delirium, especially if there is not adequate local anesthesia.

Emergence	No special considerations, except that throat packs must be removed.
Blood and fluid requirements	IV: 20–22 ga × 1 BSS @ 2–4 mL/kg/h
Monitoring	Standard monitors
Positioning	✓ and pad pressure points ✓ eyes

POSTOPERATIVE

Complications	PONV	These patients may swallow blood, with consequent PONV.
Pain management	Oral analgesics	Acetaminophen can be started ~4 h after discharge.
Recovery and discharge	In the dental office, there may not be a separate, designated recovery area. Usually, the pediatric patient is recovered by the anesthesiologist in the treatment room for at least 30–45 min, or until the patient is opening his/her eyes and maintaining a normal airway without assistance. At this point, the iv and monitors can be removed and the patient can continue recovering with his/her parents or family in the office until the patient can keep his/her eyes open, follow commands, seem calm, and is free of nausea. Postop instructions given to the parents include no outside or unsupervised activities for the rest of the day, no drinking or eating in the car on the way home, clear liquids for the first h, and	

	gradually advancing the diet to soft foods. The anesthesiologist should give the parents his/her pager or cell phone number so they can call with any questions that arise after they are at home. Regardless, a follow-up phone call from the anesthesiologist is always a good policy.

DENTAL IMPLANTS AND BONE GRAFTING

SURGICAL CONSIDERATIONS

Azeem K. Lakha, DMD

Description

A dental implant consists of a tooth root–shaped titanium post that is used to support a crown, bridge, or denture. Dental implants are inserted surgically into the mandibular or maxillary alveolar bone where teeth are missing. Single implants may be done with local anesthesia, but multiple or complex procedures are best accomplished with iv sedation. After the local anesthetic is administered, a mucoperiosteal flap is raised over the edentulous alveolus, and the bone is exposed. Precise drill holes are made in the bone, and the implants are screwed or tapped into place. Bone grafting may be necessary around the implants to fill in defects and is carried out using autologous, allogeneic, xenogeneic, or synthetic materials. In most cases, the gum tissue is closed over or around the implant. The bone is allowed to heal around the implant, and 2–6 mo later, the implant can be used to attach crowns, bridges, or dentures. In cases where there is insufficient bone, a bone graft is necessary **before** implants can be placed. Typically, bone grafts are allowed to heal for 6 mo before implant insertion. Most minor grafting procedures are accomplished in the dental office under iv sedation and local anesthesia. Major grafts requiring extraoral donor sites may have to be done under GA via nasal ETT intubation.

Usual preop diagnosis

Acquired or congenital absence of dentition

SUMMARY OF PROCEDURE	
Position	Supine, with head tilted back, using an articulating headrest in the dental chair

Incision	Intraoral mucoperiosteal flaps
Special instrumentation	Precordial stethoscope
Unique considerations	Throat packs are used to prevent aspiration of teeth, crowns, blood, and irrigation fluids. Ensure that they are removed at the end of the case.
Antibiotics	Prophylactic antibiotics, given orally before surgery
Surgical time	1–3 h
EBL	25–100 mL
Mortality	Rare
Morbidity	Infection: 1% Bleeding: Rare Delayed bleeding: Rare Aspiration: Rare
Pain score	1–4, depending on procedure

PATIENT POPULATION CHARACTERISTICS	
Age range	17+ yr
Male:Female	1:1
Incidence	Becoming more common as technology improves
Etiology	Tooth loss

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Many patients having dental implant procedures are older and/or have multiple medical problems. The anesthesiologist should be consulted in advance about these patients so that questions about their medical conditions can be answered and a current list of medications can be obtained. Sometimes, the patient's primary care physician needs to be contacted to discuss details of medical Hx. If chronic medical conditions are stable, patients often can receive "conscious sedation" and monitoring by the anesthesiologist for this procedure in the office.

Premedication: Usually not necessary

INTRAOPERATIVE

Anesthetic technique

MAC: These patients are not intubated electively. In the adult patient having dental implants, the maintenance of a lightly sedated state can be achieved using a combination of iv midazolam, fentanyl (or meperidine), ± low-dose ketamine (20–30 mg/dose). A pulse oximeter, ECG, BP cuff, precordial stethoscope or $ETCO_2$ monitor, and nasal O_2 should be in place. Glycopyrrolate (0.1–0.2 mg) should be given to decrease secretions. Dexamethasone 4–8 mg and metoclopramide 15 mg are useful as an antiemetic combination. Usually, the oral surgeon needs the patient's cooperation at some point during the procedure; therefore, propofol is not an ideal drug to use. It can be given, however, in small doses to the patient who requires more than the other drugs for sedation. Occasionally, the adult patient in this setting will require a nasal airway.

Emergence	No special considerations, except that throat packs must be removed at end of procedure.
Blood and fluid requirements	IV: 20–22 ga × 1 BSS @ 1–3 mL/kg/h
Monitoring	Standard monitors
Positioning	✓ and pad pressure points ✓ eyes

POSTOPERATIVE

Complications	PONV	These patients may swallow blood, with consequent PONV.
Pain management	Oral analgesics	
Recovery and discharge	In the dental office, there may not be a separate, designated recovery area. Usually, the patient is recovered by the anesthesiologist in the treatment room until the patient is responding appropriately. At this point, the iv and monitors can be removed and the patient can continue recovering with	

	family/accompanying adult as necessary in the office until the patient is comfortable, free of nausea, and able to drink water or fruit juice.

SECTION **15.0**

EMERGENCY PROCEDURES FOR THE ANESTHESIOLOGIST

ANESTHESIOLOGIST
Frederick G. Mihm, MD